Night and Day: Temperature and Structural Transformation in India *

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Abstract

Do rising temperatures accelerate or arrest structural transformation? Evidence from India (1981-2011) reflects two opposing implications under higher minimum (ΔT_{wmin}) vs maximum (ΔT_{wmax}) wet-bulb daily temperatures. ΔT_{wmin} increases cereal yields and demand for farm labor, pulling seasonal workers into full-time agricultural employment. In contrast, ΔT_{wmax} drives land abandonment and depresses farm wages, reducing demand for agricultural labor. Simultaneously, harvest prices increase as workers shift towards flexible but precarious urban labor markets. Over 30 years, warming nights fuel while extreme daytimes freeze structural transformation.

Keywords: Climate Change, Structural Transformation, Agriculture, India

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1 Introduction

Structural transformation is widely considered the foundation of economic development (Matsuyama 1992; Herrendorf et al. 2014) , but faces a significant challenge due to the "agricultural productivity gap" in developing economies[1]. It is unclear whether climate change – particularly rising temperatures – widens or shrinks this productivity gap. While some evidence suggests heat shocks push workers into urban sectors (Jessoe et al. 2017; Colmer 2021), other evidence finds that the resulting "food problem" traps labor in agriculture in the long run (Nath 2025; Liu et al. 2023). This paper presents an explanation for diverging empirical evidence on climate change and sectoral reallocation of labor. Relying on daily means compresses an economically meaningful heterogeneity in the labor response. Changes in the nighttime dew point (ΔT_{wmin}) have relatively greater impacts on crops, whereas farm labor faces heat stress under peak daytime temperatures (ΔT_{wmax}). The diurnal phase of heat determines the pace of structural transformation.

The debate traces to two separate bodies of evidence. On the other hand, Liu et al. (2023) find that rising temperatures reduce nonagricultural worker shares over six decades of the Indian Census, with effects strengthening over longer horizons. Their evidence points to a demand-side mechanism: agricultural productivity losses reduce local demand for nonagricultural goods and services. Nath (2025) formalizes this idea, showing that subsistence requirements create a "food problem" under a heat-induced decline in farm productivity. The non-homothetic demand for food intensifies agricultural specialization rather than reallocating labor into non-agricultural sector. The two results are not obviously reconcilable: both use temperature variation and microdata in India, credible identification, and examine labor reallocation. They simply disagree on the direction.

Our mechanism is physiological and grounded on the following assumption: human labor is outdoors during the daytime, but crops are exposed 24/7. Therefore, temperature changes during the dew point (ΔT_{wmin}) of a day have higher impacts on land relative to labor. The daytime peak temperature (T_{wmax}) impacts land and labor productivity. Aggregate metrics of temperature – such as degree days, daily means, or monthly averages – pool two physiologically distinct phenomena. The biological stress that accumulates with warming nights, and the physical incapacitation resulting from peak daytime exposure. Night and daytime temperatures do not necessarily move together. Using approximated wet-bulb temperatures, I decompose heat exposure into the nighttime (T_{wmin}) and daytime (T_{wmax}) components of daily temperature. This bifurcation reveals opposing long-run impacts on structural transformation.

I establish this mechanism using three data sources at the district and town level in India between 1981-2011. The primary source is the Census of India, which I use to construct a balanced panel of approximately 300 districts and 3,000 towns across four decadal rounds, harmonized to their administrative boundaries in 2011. Agricultural outcomes are provided by ICRISAT at the

[1] In India, agriculture employs 42% of the labor force and contributes to 16.3% of GDP (World Bank, 2025)

district level, covering area, production, yield and harvest prices annually for key crops of India (1966-2020). Data released by Ruggles et al. (2024) provides evidence on household employment from three rounds of the National Sample Survey (1987, 1999, 2009). The data compilation harmonizes evidence from population and household surveys. Daily gridded weather data obtained from the Indian Meteorological Department are interpolated to district and town-level climate. Daily observations are averaged over 10-year growing seasons to capture climate exposure most relevant for structural transformation decisions. Town and district-level climate is constructed by inverse-distance weightings based on the unit's geographic centroid.

The findings proceed in three steps. At the district level, I first establish that degree days, a standard aggregate measure, masks a diverging decomposition. Nighttime warming raises agricultural worker shares and cereal production, with no significant effect on cultivated area. This is consistent with Stone-Geary labor hoarding: as crops face biological stress, the subsistence constraint binds, and households intensify agricultural effort rather than reducing it. Daytime extremes operate differently. They reduce cultivated area and farm labor while raising harvest prices — a pattern consistent with a supply shock, where agricultural land-use adapts or perishes. The F-statistic rejecting equality of the two temperature coefficients on the share of agricultural workers in districts is 9.04, entirely driven by agricultural labor rather than cultivation.

The second step moves to towns – a unit of analysis which, to my knowledge, has not been utilized to study climate and structural transformation. The second step utilizes a novel town-level panel with consistent boundaries of Census-designated urban areas. Towns are the spatial destination for structural transformation, housing the non-agricultural economy to absorb distressed agricultural workers. I show that, in the short run, daytime heat increases the share of marginal/seasonal work, employed fewer than 183 calendar days preceding the Census (1981-2011). NSS urban households (1989-2009) rellocate into the lower end of non-agricultural occupations such as trade, transport, and domestic work.

Over a longer horizon, the consequences are more severe. The baseline trajectory of these towns over 30 years is a transition from agricultural to marginal employment: the mean long-difference in share of agricultural workers is -0.08 and marginal workers is 0.07. Persistent daytime extremes lead to a secular halt of reallocation across the remaining sectors, "freezing" structural transformation. Nighttime warming allocates workers into agriculture over 10 years, while they reallocate to nonagricultural employment over 30 years. However, the short-run shift away from agricultural cultivation to wage labor reflects a compositional degradation of the workforce rather than a genuine sectoral transition.

The third step uses NSS household data to probe the micro-mechanisms underlying these aggregate patterns. Nighttime warming generates a significant increase in agricultural wage employment in urban areas, corroborating the cultivator-to-laborer shift visible in Census data and confirming its presence in an independent survey instrument. Daytime extremes drive exit from outdoor manual work — agriculture, construction, and mining — and reallocation toward urban nonagricultural employment, but without generating formal sector wage growth. This pattern rules out a simple labor market flooding story: self-employment in trade does not spike under daytime heat, and trade wages do not fall. Instead, the evidence is consistent with a physical displacement from heat-exposed work into shaded or flexible employment — a survival response, not a productivity-driven transition.

These findings speak to several strands of existing work. The paper advances Liu et al. (2023) along two dimensions: it introduces the minimum vs maximum decomposition that their specification implicitly aggregates, and it operates at the town level rather than the district level. By doing so, it isolates the specific urban destinations for distressed labor. The nighttime mechanism complements the food problem under Nath (2025): both models generate agricultural subsistence trap, but the mechanism here is agro-physiological rather than trade-structural. The daytime mechanism qualifies results from Colmer (2021): workers do exit agriculture under peak heat, but the receiving formal nonagricultural sector is productivity constrained, as Somanathan et al. (2021) establish for Indian manufacturing. While the agricultural exit occurs, the non-agriculural absorption does not. Together, they reveal the Indian transition to services not as an engine of growth, but as a physiologically constrained churn for survival.

The remainder of the paper proceeds as follows. Section 1 develops the conceptual framework and mechanism. Section 2 describes the data and empirical strategy. Section 3 reports results from districts, towns and urban households. Section 4 concludes.

2 Conceptual Framework

A wide array of literature documents the impacts of heat are significantly higher on agriculture relative to nonagriculture. [citation needed] While the effects of temperature on structural transformation are inherently nonlinear, extreme heat in particular distresses labor and farm productivity in this region (Welch et al. (2010), Peng et al. (2004), Dunne et al. (2013), Somanathan et al. (2021)). A growing literature has identified the effects of agricultural productivity on labor reallocation. While an improvement in farm productivity increases demand for the nonagricultural sector (**Burke2018**) , a reduction Findings from Liu et al. (2023) indicate that rising temperatures reduce productivity depressing demand for the non-farm sector in the long run (Liu et al. (2023), Nath (2025)).

I establish two empirical facts in the Indian context: warming (ΔT_{wmin}) improves cereal yields, while extremes (ΔT_{wmax}) reduce farm acreage. These serve as the two central assumptions of the model, consistent with [citation needed]. To maintain tractability, the model abstracts from several spatial and temporal channel, each of which have significant nonlinear implications on structural transformation. A growing literature documents how roads, migration, infrastructure and trade individually influence structural transformation. [citation needed].

The framework involves a simple general equilibrium under a closed regional economy with two sectors: agriculture and non-agriculture $\{Y_a, Y_n\}$ Our model incorporates underlying mechanisms to prior models of structural transformation from Matsuyama (1992), Gollin and Rogerson (2014), Bustos et al. (2016) and Liu et al. (2023). The main focus is to model the implications for structural transformation under a Hicks-neutral shock to productivity $A_a(\Delta T_{wmin})$ vs biased shock to land $Z_a(\Delta T_{wmax})$ in the agricultural sector. We show how the nonagricultural sector can expand or shrink given either shock. aggregate supply satisfies aggregate demand locally. First, the temperature shocks are Hicks-neutral in non-agricultural productivity,

In the nonormalize $Z_n = 1$.

in agriculture and nonagriculture, consumed by households $\{c_a, c_n\}$. Production in sector $i \in \{a, n\}$ follows the functional form $Y_i = A \cdot F(H, Z)$ demands labor hours (H) and land (Z) with total factor productivity A. Labor hours $\{H_a, H_n\}$ consists of formal employment $\{L_a, L_n, L_s\}$. Labor L_a and L_n represents formal employment in either agriculture and non-agriculture, respectively; while L_s captures seasonal work providing employment at a flexible wage.

2.1 The Producer Problem

The representative household is endowed with land Z and selects an occupation L_i such that $L_a + L_n + L_s = L$ strictly holds. Labor in the seasonal occupation (L_s) serves as a multiplier to labor in either Agriculture (L_a) and Non-Agriculture (L_n). Agricultural production faces a physical capacity constraint where land Z_a determines the level of output.

Employment L_s serves as part-time labor towards production in sector a or n. This results in effective labor $H_i = (1 + \lambda_i)L_i$, where formal workers supply one unit of labor and informal workers supply a multiplier $\lambda_i \in [0, 1]$. We define $s_i = L_{s,i}/L_i$ as the ratio of informal seasonal workers attached to formal sector $i \in \{a, n\}$.

Crucially, nighttime warming T_{wmin} acts as a Hicks-neutral productivity shock to agriculture $\mathcal{A}_a$, while daytime extreme heat T_{wmax} reduces human physiological capacity in non-agriculture $\mathcal{A}_n$ and physically damages the viable land supply Z in agriculture.

$$Y_a = \mathcal{A}_a(T_{wmin}) \cdot Z(T_{wmax})^{\theta_2} \cdot (L_a \lambda_a)^{\theta_1}$$

$$Y_n = \mathcal{A}_n(T_{wmin}, T_{wmax}) \cdot (L_n \lambda_n)^{\eta}$$

To maintain log-linear tractability in the dual-labor market, we apply the standard first-order Taylor approximation for small ratios, $\ln(1 + s_i) \approx s_i$. Taking the natural logarithm of the agricultural production function yields a strictly additive system.

$$\ln Y_a \approx \ln \mathcal{A}_a(T_{wmin}) + \theta_2 \ln Z(T_{wmax}) + \theta_1 \ln L_a + \theta_1 s_a$$

2.2 The Consumer Problem and General Equilibrium Prices

The representative household maximizes Stone-Geary utility over agricultural (c_a) and non-agricultural (c_n) consumption, subject to a biological subsistence constraint $\bar{a}$.

$$U(c_a, c_n) = \alpha \ln(c_a - \bar{a}) + (1 - \alpha) \ln(c_n)$$

In a closed regional economy facing spatial trade frictions, local markets must clear such that aggregate demand equals local supply. Setting the non-agricultural good as the numeraire ($p_n = 1$), the aggregation of individual household budget constraints yields the general equilibrium relative price for the agricultural good p_a.

$$p_a = \frac{\alpha}{1 - \alpha} \frac{Y_n}{Y_a - \bar{a}L}$$

By differentiating the natural logarithm of this price with respect to temperature T, we establish the General Equilibrium price response. Letting ϵ_x denote the temperature semi-elasticity $\frac{d \ln x}{dT}$, the relative price shock is mathematically determined by the difference in sectoral elasticities.

$$\epsilon_{pa} = \epsilon_{Yn} - \mathcal{M} \epsilon_{Ya}$$

Here, $\mathcal{M} = \frac{Y_a}{Y_a - \bar{a}L}$ is defined as the subsistence multiplier. Because humans must consume caloric minimums to physically survive, $Y_a > Y_a - \bar{a}L$ structurally guarantees that the multiplier strictly exceeds one ($\mathcal{M} > 1$). This multiplier dictates that physical shocks to agricultural output will rigidly dominate local price dynamics compared to equivalent shocks in the non-agricultural sector.

2.3 The Producer Problem and Formal Labor Demand

Competitive firms in both formal sectors hire labor until the nominal wage equals the value of the marginal product. Because the informal sector operates merely as a structural multiplier on output, the formal agricultural wage w_a is determined cleanly by the marginal product of the formal worker L_a.

$$w_a = p_a \frac{\partial Y_a}{\partial L_a} = p_a \theta_1 \frac{Y_a}{L_a}$$

Rearranging this first-order condition provides the formal agricultural labor demand.

$$L_a = \theta_1 \frac{p_a Y_a}{w_a}$$

Log-differentiating with respect to temperature yields the structural elasticity of agricultural labor demand.

$$\epsilon_{La} = \epsilon_{pa} + \epsilon_{Ya} - \epsilon_{wa}$$

2.4 Occupational Sorting and the Seasonal Sink

To strictly map the firm's multiplicative elasticity (ϵ_{La}) to the additive linear shares observed in the census data (l_a), we apply the chain rule $\frac{dl_a}{dT} = l_a \cdot \epsilon_{La}$. Because the share of formal workers is strictly positive ($l_a > 0$), the absolute change in the labor share perfectly preserves the theoretical sign of the elasticity. Workers who are not absorbed by formal agriculture or formal non-agriculture fall into the residual seasonal sink, clearing the additive labor market identity.

$$\frac{dl_s}{dT} = -\left(\frac{dl_a}{dT} + \frac{dl_n}{dT}\right)$$

Seasonal workers face a flexible, precarious wage w_s that must satisfy the biological survival inequality $w_s \geq p_a \bar{a}$. This closed system yields diverging, structurally bounded comparative statics for the diurnal temperature cycle. Under daytime extreme heat (T_{wmax}), the severe destruction of viable land ($\epsilon_Z \ll 0$) collapses physical agricultural output. The subsistence multiplier translates this extensive margin scarcity into a violent price spike ($\epsilon_{pa} \gg 0$), which forces households to exhaust their budgets entirely on food, crushing local demand for non-agricultural goods. Consequently, formal non-agriculture becomes stagnant and cannot expand ($\frac{dl_n}{dT} \approx 0$). Simultaneously, the physical destruction of the farm crashes the marginal product of labor, depressing formal wages and forcing a massive farm exit ($\epsilon_{La} < 0 \implies \frac{dl_a}{dT} < 0$). With the formal agricultural sector contracting and the non-agricultural sector starved of demand, displaced workers are structurally forced into informal precarity, generating an explosion of the seasonal sink ($\frac{dl_s}{dT} > 0$). Conversely, under nighttime warming (T_{wmin}), the intensive margin agricultural productivity boom ($\epsilon_{\mathcal{A}a} > 0$) expands output, lowering local food prices and stabilizing baseline non-agricultural demand. The physical productivity surge keeps agricultural wages lucrative, aggressively pulling workers into the formal farm sector ($\frac{dl_a}{dT} > 0$) and directly draining the informal seasonal pool ($\frac{dl_s}{dT} < 0$).

3 Data and Empirical Strategy

3.1 Data Sources

I harmonize four datasets to link climate, agricultural production and structural transformation: (1) daily gridded weather data from the IMD, (2) district agricultural (ICRISAT), (3) district and town level tables from the Indian Census, and (4) urban household-level employment data from the NSS. These datasets span across 4 decades of the (Census, IMD, ICRISAT), with a shorter window from the NSS (1987,1999,2009).

Indian Meteorological Department Gridded data were obtained from the Indian Meteorological Department (IMD) on air temperature (1° resolution) and precipitation (0.25° resolution) between 1951-2023. Consistent with the approach in Liu et al. (2023), daily observations are averaged over growing seasons of the past 10 years. The growing season marks the months of June-February, which captures cropping cycles of rice in the monsoon (kharif) and wheat in the winter (rabi) seasons. Daily heat exposure is measured through Degree Days (DDs), studied under Blakeslee and Fishman (2015) in India and Schlenker et al. (2006) in the US. Degree days rescale and sum up agriculturally relevant temperatures between 8-32 C.

$$DDs = \sum_d D(\mathrm{T}_d) \text{ ; where } \mathrm{T}_d = \frac{\mathrm{T}_{min,d} + \mathrm{T}_{max,d}}{2} \tag{1}$$

$$D(\mathrm{T}) = \begin{cases} 0, & \text{if } \mathrm{T} \leq 8°\mathrm{C} \\ \mathrm{T} - 8, & \text{if } 8°\mathrm{C} < \mathrm{T} \leq 32°\mathrm{C} \\ 24, & \text{if } \mathrm{T} > 32°\mathrm{C} \end{cases} \tag{2}$$

Census of India Our core dataset is sourced from the District Census Handbooks (DCHBs) of India. Specifically, I digitize data tables from the Primary Census Abstract (PCA), covering the universe of Indian towns from 1981 to 2011. A Census town or urban area is defined as meeting 3 criteria: (1) a minimum population of 4,000, (2) at least 75% of male working population in non agricultural activity, and (3) a population density of over 400 persons per sq. km. I construct a panel of 3000 towns that appear consistently across four decades, ensuring that our analysis considers intensive margin adjustments rather than the reclassification of urban areas. For consistency in definition, occupations are aggregated to 4 mutually exclusive categories: cultivators (landowners), agricultural laborers (wage workers), non-agricultural workers and marginal workers. While the "marginal" classification prevents interpretation as either agriculture or non-agriculture, this definition permits us to measure the contractual *quality* of work.

ICRISAT The ICRISAT District Level Database (DLD) tracks agricultural inputs and production in India between 1966-present. I harmonize districts to a 2011-Census consistent boundaries, constructing an annual panel of cultivated area, production, yields and harvest prices for key grains: rice, wheat, maize, sorghum, barley and millets. I construct the mean of these measures over 10 years preceding each Census round, e.g over the years of 1971-1980 for the 1981 Census. Thus, the results from the temperature regressions compare contemporaneous horizons of 10-year windows with climate data.

National Sample Surveys While the Census provides the volume of labor reallocation, it lacks granularity on the destination sectors within non-agriculture in 1981 definitions. To identify where displaced labor ends up, I study a panel of five rounds of the National Sample Survey (NSS) Employment and Unemployment Schedules (1987-2004), as harmonized by IPUMS (Ruggles et al. 2024). I categorize occupations into "Outdoor" (Cultivation, Agricultural Labor, Construction, Mining) and "Indoor" (Manufacturing, Services) work. I further restrict our sample to urban households, serving as a benchmark to the Census town-level data. This linkage allows us to observe the destination for heat-displaced labor, sorting into seasonal gigs rather than stable employment in manufacturing or services.

3.2 Empirical Strategy

The identification relies on two approaches: a fixed effects panel estimation (3), and a long differences approach (4) described under Dell et al. (2014), Burke and Emerick (2016) and Liu et al. (2023). Let Y_{irt} be the outcome of interest (e.g., share of workers in agriculture) for district or town i, within state r, observed at time t. $\mathbf{T}_{it}$ denotes a vector of temperature measures, defined as either degree days $\mathbf{T} = \{DDs\}$ or diurnal wet-bulb temperatures $\mathbf{T} = \{T_{wmin}, T_{wmax}\}$. Under all specifications, we control for total precipitation $\mathbf{P}_{it}$ and fit a state-specific linear time trend.

$$Y_{irt} = \beta \mathbf{T}_{irt} + \phi \mathbf{P}_{irt} + \mu_i + \lambda_t + \gamma_r \cdot t + \varepsilon_{irt} \tag{3}$$

$$\Delta Y_{ir} = \beta \Delta \mathbf{T}_{ir} + \phi \Delta \mathbf{P}_{ir} + \gamma_r + \varepsilon_{ir} \tag{4}$$

For temporal parity, we construct gridded temperature and rainfall measures as a moving average during growing seasons of the past 10 years. Decadal climate is constructed by weighting the inverse-distance from the district or town's centroid to gridpoints within a 100 km radius.

Equation 3 uses a town-decade panel with two-way fixed effects fitting a state-specific time trend. We interpret these as *short-run* (10-year) effects of heat shocks on structural transformation. Equation 4 averages the outcome Y and covariates X over two periods t_1 and t_2, regressing $\Delta Y = Y_2 - Y_1$ on $\Delta X = X_2 - X_1$. This "long-differenced" estimator recovers the effect of an increase in $\mathbf{T}$ on the outcome trend ΔY, relative to its secular trend $\Delta \bar{Y}$ over the entire sample. By sweeping out unobserved state-level trajectories $(\gamma_r \cdot t)$, we interpret these coefficients as the *long-run* (30-year) localized adaptation response of labor under sustained heat shocks.

We simultaneously regress T_{wmin} and T_{wmax} and interpret the coefficients as the local average treatment effect (LATE) of either heat shock (e.g. ΔT_{wmin}), conditional on the other end of the diurnal range (e.g. ΔT_{wmax}) being constant. We consider several alternate specifications, including spatially corrected errors (Conley 1999). Appendix table 4 reports rejection rates for the null hypothesis, regressing the true data against simulated noise. We select our preferred estimator for equation 3 as the specification that rejects the null hypothesis in approximately 5% of simulations.

This simultaneous estimator addresses two key identification challenges. First, given the high the correlation between T_{wmin} and T_{wmax} ($\rho = 0.93$), an individual specification of either measure would suffer from omitted variable bias. The simultaneous specification isolates their partial effects, effectively identifying a contraction vs expansion of the local diurnal temperature range. Second, absorbing state-level trends ensures we account for time-varying institutional policies by Indian states. Additionally, this ensures we compare towns within their local economic geography, under similar baseline trends in climate. Correlograms of our residuals confirm that the spatial correlation of these heat shocks on key outcomes decays within a 100 km radius.

4 Results

Table 1 establishes the empirical fact on the production and prices of key cereals in India. : rice, wheat, maize, sorghum, barley and millets. These crops dictate the baseline for farm income managed through government procurement in India. The table presents estimates of the degree days (DDs) and wet-bulb diurnal decomposition T_{wmin}, T_{wmax} on the agricultural economy in districts. The primary result from this paper, that the diurnal heat cycle governs dual mechanisms, is supported by the structure of Table 1 alone.

Panel A shows that an increase in DDs negatively impacts the production of cereals, reducing cultivated area and increasing harvest prices. Yet, the resulting effect on agricultural worker shares is contradictory: a decrease in cultivators by 3.8 p.p ($p < 0.05$) and increase in laborers by 3.2 p.p ($p < 0.05$). If production is falling and land is abandoned, the agricultural sector should not absorb wage labor. Panel B reveals the aggregate metric of degree days suppresses two opposing economic mechanisms. The nighttime (T_{wmin}) and daytime (T_{wmax}) decomposition results in estimates with opposing signs across almost every outcome. The F-statistic rejects their equality ($\beta_{wmin} = \beta_{wmax}$) ranges from $F = 12.01$ on cereal production to $F = 23.83$ on harvest prices. However, the coefficients T_{wmin} and T_{wmax} on the share of cultivators do not significantly differ $F = 0.12$, suggesting the channel operates through agricultural labor rather than self-owned cultivation.

Moderate heat measured with T_{wmin} supports the commercialization of agriculture within districts. Warming leads to a 21.7 percent reduction in harvest prices ($p < 0.01$), associated with a 27.4 percent increase in cereal production ($p < 0.05$). However, there is no detectable change in cultivated area, combined with an increase in agricultural labor shares. a shift in employment shares: cultivators decrease by 3.8 p.p ($p < 0.05$) and agricultural laborers increase by 3.2 p.p ($p < 0.05$) The evidence indicates no negative impacts of moderate warming on farm productivity, rather an increase in the opportunity cost for farming under falling prices. This shift from self-cultivation to

Table 1: Long-Run Effects of Temperature on Agricultural Production and Sectoral Reallocation in Districts

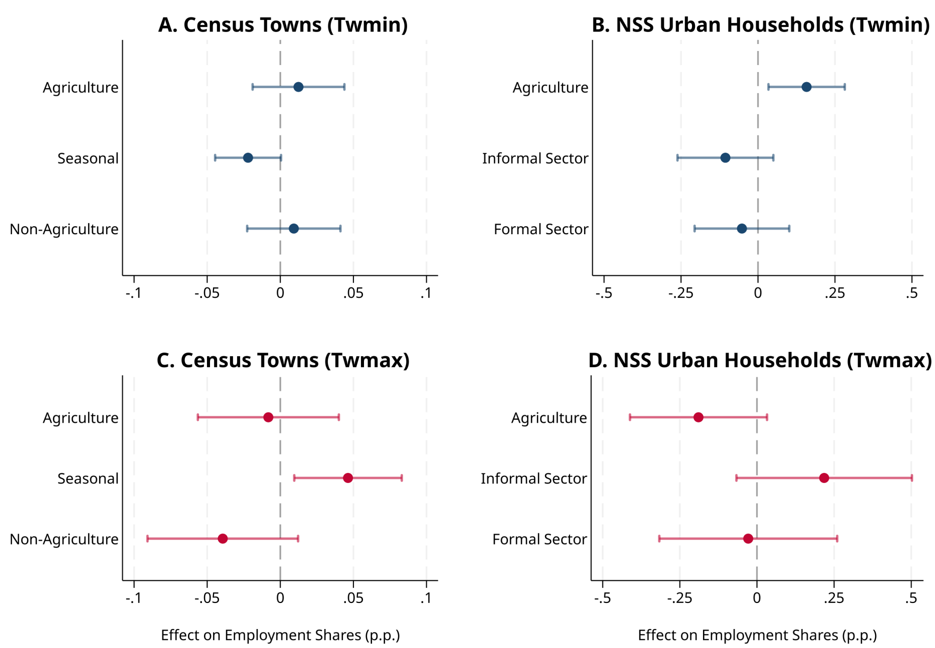
Panel A. ICRISAT Districts (1981-2011)				
	Log Levels			
	Price Index	Yield Index	Field Wage	Cultivated Area
	(1)	(2)	(3)	(4)
$\Delta\ T_{wmin}$	-0.289***	0.174**	0.195	0.119
	(0.072)	(0.086)	(0.143)	(0.083)
$\Delta\ T_{wmax}$	0.452***	-0.210	-0.455**	-0.706***
	(0.115)	(0.140)	(0.224)	(0.163)
Observations	1,191	1,211	1,115	1,243
Sample Mean	5.66	7.14	2.86	5.45
F-stat ($T_{wmin} = T_{wmax}$)	16.93	3.10	3.33	12.22
Panel B. Census Districts (1981-2011)				
	Share of Workers			
	Cultivation	Ag. Labor	Seasonal	Non-Agriculture
	(1)	(2)	(3)	(4)
$\Delta\ T_{wmin}$	-0.010	0.082***	-0.076***	0.006
	(0.023)	(0.017)	(0.020)	(0.021)
$\Delta\ T_{wmax}$	-0.029	-0.078***	0.088***	0.016
	(0.034)	(0.025)	(0.033)	(0.035)
Observations	1,186	1,186	1,186	1,186
Sample Mean	0.32	0.18	0.17	0.32
F-stat ($T_{wmin} = T_{wmax}$)	0.12	16.15	10.27	0.03
Rainfall Control	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y

Notes: Panel A measures prices, yields index for key cereals in India: rice, wheat, barley, maize, millets, sorghum. Panel B measures the district share of agricultural workers out of total workers (Census). Temperatures constructed as an average of the past decade's growing seasons (June-Feb). All specifications control for total precipitation in meters over the same period. The data is a decadal panel (1981-2011), using 2011 Census-consistent district boundaries. Additional controls include district and decade two-way fixed effects, fitting a state $\times$ decade linear trend. Standard errors clustered at the district level.

hired labor is indicative of Lewisian structural transformation, paving the transition into nonagricultural employment.

In contrast, a T_{wmax} shock is purely destructive for cereals: wiping production by 97 log points $(p < 0.01)$, cultivated area by 70 log points ($p < 0.01$), and increasing harvest prices by 42 log points $(p < 0.01)$. This collapse is associated with a reduction the share of agricultural laborers by 7.8 percentage points $(p < 0.01)$ in districts. This is indicative of a contraction in the agricultural sector: peak daytime temperatures damage crops and render sustained outdoor labor infeasible. Land is abandoned because it cannot be physically worked.

Figure 1: 10-Year Effects of Temperature on Urban Employment Shares



Notes: The figure plots coefficients with 95% C.I. of a 10-year shock to T_{wmin} vs T_{wmax} on the share of households employed per sector (Census 1981-2011; NSS 1987-2009). Temperatures constructed as a moving average during growing seasons (June-Feb) of the decade preceding observation. All specifications control for total precipitation in meters over the same period. Additional controls include district and decade two-way fixed effects, fitting a state $\times$ decade linear trend. Standard errors clustered at the district level.

Figure 1 focuses on the effects of decadal shocks to T_{wmin} and T_{wmax} on the share of employment in Census (1981-2011) and National Sample Survey (1987, 1999, 2009). The independence of these enumeration exercises and overlap in their time horizon permits cross comparability and a finer disaggregation of occupations. These data captures the share of employment within towns (Census) and within districts (NSS urban households).

In the short-run panel, the nighttime shock reshapes the composition of agricultural work within towns without triggering a sectoral exit. T_{wmin} raises the share of agricultural laborers (wage workers) by 2.4 percentage points ($p < 0.10$) while reducing the share of cultivators by 2.1 percentage points ($p < 0.05$). These effects are statistically indistinguishable in magnitude and opposite in sign, producing a near-zero net effect on total agricultural employment. The F-statistic rejecting their equality for the combined agricultural share is 3.18. This within-agriculture shift from cultivation to wage labor is consistent with the district-level evidence: as yields stagnate and prices fall, the farm becomes less viable as an enterprise and more viable only as a source of wage income. The cultivator, facing declining returns to land ownership, enters the wage labor market without leaving the sector.

The daytime shock produces a different short-run pattern. T_{wmax} reduces the share of other non-agricultural workers by 4.8 percentage points (p ¡ 0.10) with a positive but marginally significant effect on seasonal employment (+3.0 percentage points, p = 0.10). This suggests that peak daytime heat immediately erodes the formal non-agricultural workforce and pushes workers toward casual and seasonal arrangements — the first sign of the long-run freeze that the long-differenced estimates confirm.

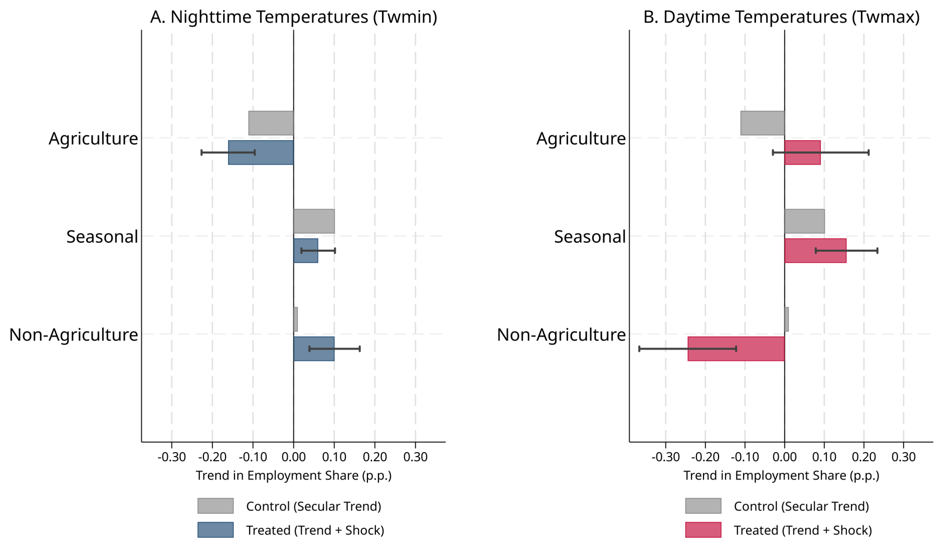
The nighttime panel (Panel A) shows a consistent pattern of economically small and statistically imprecise effects across most sectors, with two notable exceptions. Agricultural labor and cultivation move in the same direction as the Census evidence, though precision is lower given the NSS sample size. Formal non-agricultural sectors in manufacturing and high services show coefficients close to zero – together with the Census town findings, they suggest warming does not erode formal nonagricultural employment in the short run.

The daytime panel (Panel B) reveals the mechanism underlying the long-run non-agricultural collapse: an exit from outdoor manual labor in the primary sectors of agriculture, construction and mining. Distressed labor reallocates towards low services of trade, transport, hospitality and domestic work seeking flexible and partially shaded employment. Higher value non-agricultural work in manufacturing, high services coefficients are indistinguishable from zero.

Table 5 and Figure 1 together suggest that workers displaced from outdoor sectors by daytime heat do not access formal employment; instead, they cycle through informal arrangements in the lower end of urban services. The town's formal economy does not absorb them.

Figure 2 shows the 30-year consequences of persistent rising heat. The baseline trajectory of towns in the sample is to shed agricultural employment: the mean long-difference in agricultural worker share is -0.11, reflecting three decades of secular structural change. Against this baseline, the long-run effect of T_{wmax} on agricultural employment is +0.13 ($p < 0.01$) — a near-perfect 'halt' of the natural transformation process. The corresponding long-run non-agricultural effect is -0.15 ($p < 0.01$): the formal non-agricultural sector, rather than expanding to absorb displaced rural workers, contracts. Taken together, these estimates imply that daytime heat simultaneously prevents exit from agriculture and forecloses entry into non-agriculture — a structural arrest rather than a reallocation.

Figure 2: 30-Year Effects of Temperature on Structural Transformation in Census Towns



Notes: The figure plots coefficients with 95% C.I. of a min vs max wet-bulb temperature shock against the secular trend on worker shares. Data is a long-differenced panel of Census towns subtracting the averages of period 1 (1981-91) from period 2 (2001-11). Estimates control for rainfall and state fixed effects. Standard errors clustered at the district level.

The long-run nighttime effect tells a different story. T_{wmin} generates a significant positive effect on the non-agricultural worker share (+0.07, $p < 0.05$) over 30 years. At first glance, this is not inconsistent with the short-run evidence. However, they reflect how the 10-year within-agriculture churn accumulates into a partial sectoral transition. The mechanism is a distress-driven exit from agriculture and seasonal work, as persistently depressed agricultural incomes make farming. Work shifts toward nonagricultural employment and settles into low-end services, marking stunted structural transformation.

5 Conclusion

This paper synthesizes recent empirical evidence of temperature on employment and local economic activity. The effects are modelled as linear estimation of rising minimum vs maximum temperatures in a Matsuyama (1992) closed economy. The approach has a few limitations. Recent empirical literature finds significant nonlinearities on effects of temperature across time and space. This evidence is also supported by the paper's underlying dataset, where several outcomes are discontinuously correlated with T_{wmin} and T_{wmax}. Further, it is not directly obvious whether harvest prices would rise or fall under an open economy. Fully modeling these thresholds and the associated spatial sorting is beyond the scope of this paper's linear framework.

Keeping the limitations in mind, the evidence presented is a consistent divergence in economic incentives from the warming night and scorching day. Holding the other constant, warming T_{wmin} benefits agricultural productivity: shifting self-cultivation to hired wage labor, which enables a transition to non-agriculture in the long run. Daytime T_{wmax} damages farms and drives urban labor precarity. In the long-run, persistent daytime heat halts structural transformation in towns, facing reduced demand for formal non-agriculture. These findings indicate that the Indian urban transition to informal services reflects a physiologically constrained churn for survival.

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6 Appendix

Table 2: Towns Summary

	1981	1991	2001	2011
Panel A: Municipal Towns				
Town Characteristics				
Population	55,251.21	106098.98	150550.17	202954.42
Total Workers	16,203.97	31,778.95	47,476.02	71,007.33
Area (sq. km)	16.58	21.66	29.56	31.82
New Towns (0/1)	0.12	0.08	0.04	0.14
Decommissioned Towns (0/1)	0.00	0.00	0.00	0.00
Share of Workers by Category				
Non-Agricultural	0.75	0.75	0.80	0.79
Agricultural Labor	0.11	0.11	0.05	0.06
Cultivation	0.10	0.10	0.04	0.04
Seasonal	0.04	0.03	0.11	0.12
Climatic Indicators (Growing Seasons)				
Degree Days (DDs)	15.75	15.83	15.96	16.55
Daily Temp (° C)	23.81	23.91	24.04	24.61
Nighttime Temp: Tmin (° C)	17.91	17.90	18.18	18.75
Daytime Temp: T_max (° C)	29.71	29.92	29.90	30.47
Daily Precipitation (mm.)	3.12	3.11	3.05	3.00
Observations	1,838	2,044	1,798	1,817

Notes: Summary statistics on towns in sample using data from the District Census Handbook (1981-2011).

Table 3: District Summary

	1981	1991	2001	2011
Panel A: Rural				
Characteristics				
Population (in Thousands)	1,476.74	1,765.70	1,574.63	1,683.07
Total Workers (in Thousands)	580.97	721.67	843.99	920.10
Share of Workers by Category				
Non-Agricultural	0.18	0.19	0.39	0.39
Agricultural Labor	0.23	0.25	0.14	0.17
Cultivation	0.47	0.44	0.27	0.22
Seasonal	0.11	0.12	0.20	0.21
Climatic Indicators (Growing Seasons)				
Degree Days (DDs)	15.64	15.76	15.78	16.05
Daily Temp (° C)	23.70	23.83	23.85	24.12
Nighttime Temp: Tmin (° C)	17.80	17.89	18.00	18.24
Daytime Temp: T_max (° C)	29.60	29.76	29.70	29.99
Daily Precipitation (mm.)	3.18	3.18	3.18	3.06
Observations	286	286	286	286
Panel B: Urban				
Characteristics				
Population (in Thousands)	465.02	628.86	628.64	817.22
Total Workers (in Thousands)	138.70	189.45	220.20	323.46
Town Changes				
Total Towns	11.44	11.94	10.86	12.06
New Towns	3.30	0.96	0.84	1.58
Decommissioned Towns	0.44	0.46	1.93	0.38
Share of Workers by Category				
Non-Agricultural	0.81	0.81	0.83	0.80
Agricultural Labor	0.07	0.09	0.03	0.04
Cultivation	0.08	0.08	0.04	0.03
Seasonal	0.03	0.03	0.10	0.12
Climatic Indicators (Growing Seasons)				
Degree Days (DDs)	15.64	15.76	15.78	16.05
Daily Temp (° C)	23.70	23.83	23.85	24.12
Nighttime Temp: Tmin (° C)	17.80	17.89	18.00	18.24
Daytime Temp: T_max (° C)	29.60	29.76	29.70	29.99
Daily Precipitation (mm.)	3.18	3.18	3.18	3.06
Observations	286	286	286	286

Notes: Summary statistics on districts in sample using data from the District Census Handbook (1981-2011).

Figure 3: Thermal Zones and Physiological Limits

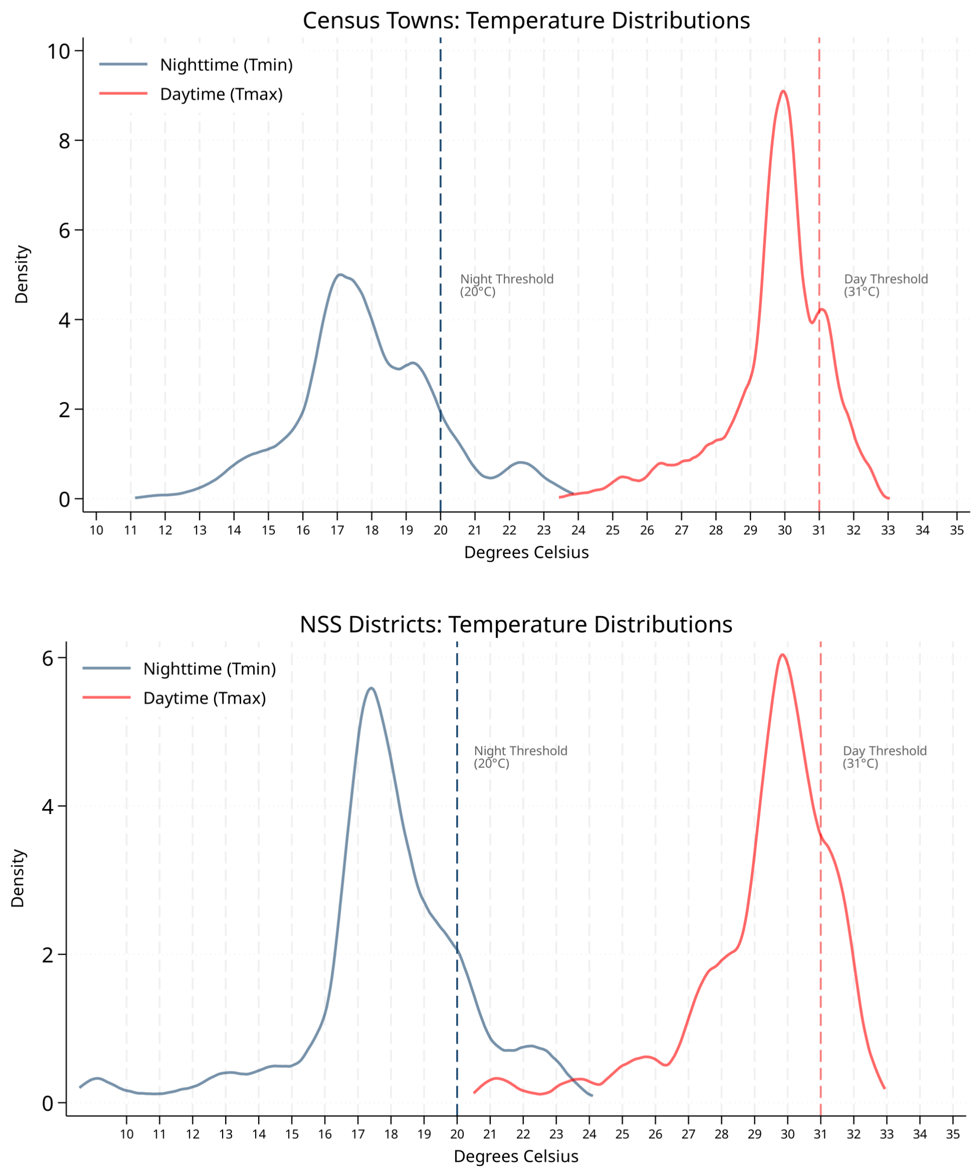


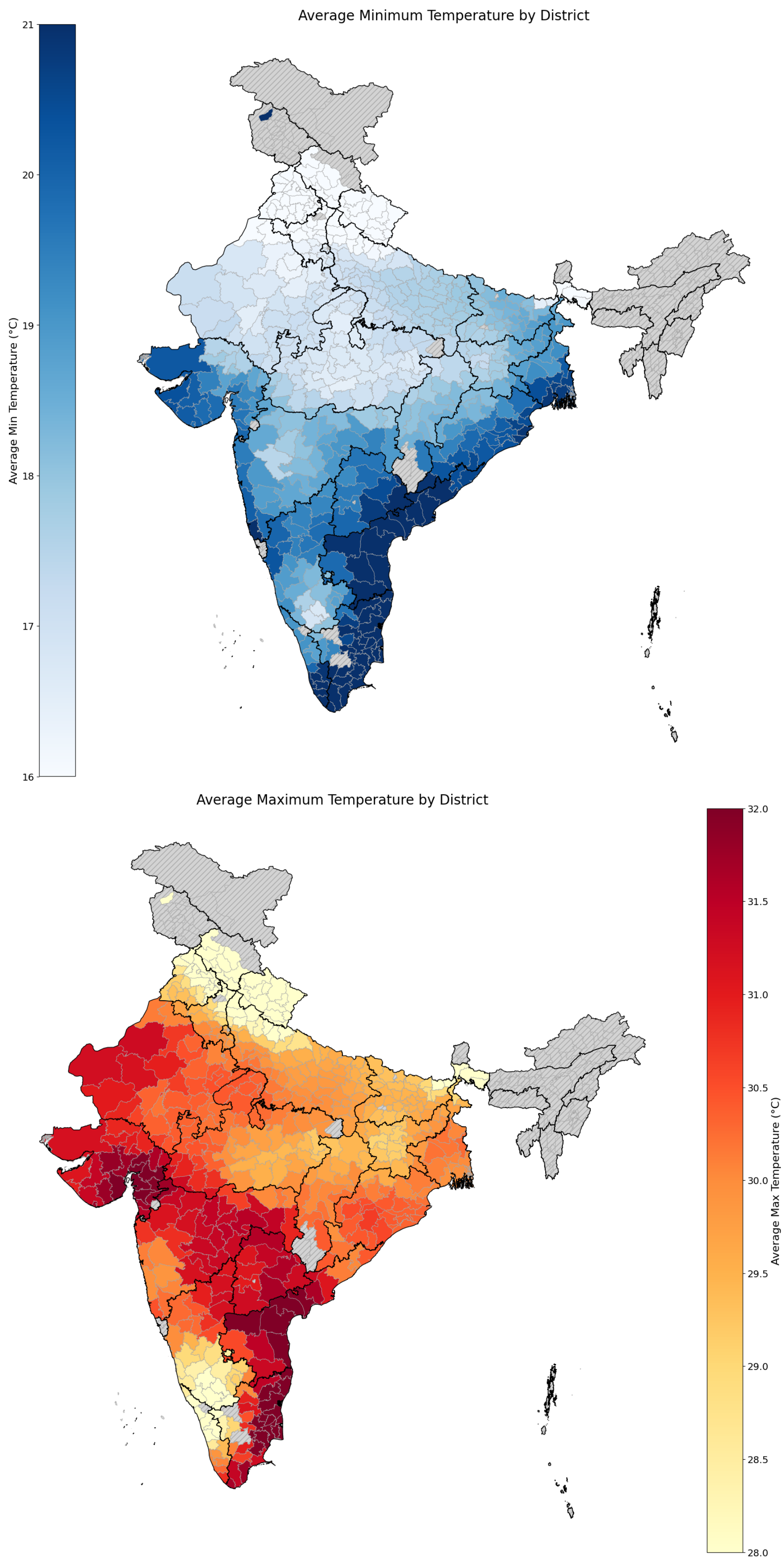
Table 4: Monte Carlo Rejection Rates on Town Data

	B1 Rej. Rate		
	Ag. Share	Non-Ag. Share	Seasonal Share
rej_B1_unadj	0.052	0.064	0.046
rej_B1_rob	0.049	0.062	0.046
rej_B1_CLdt	0.055	0.063	0.045
rej_B1_CLdtTRdt	0.052	0.061	0.041
rej_B1_CLdtTRst	0.060	0.065	0.043
rej_B1_con1	0.060	0.065	0.055
rej_B1_con2	0.069	0.066	0.059
Simulations	1000	1000	1000

	B2 Rej. Rate		
	Ag. Share	Non-Ag. Share	Seasonal Share
rej_B2_unadj	0.055	0.048	0.048
rej_B2_rob	0.053	0.048	0.049
rej_B2_CLdt	0.054	0.051	0.049
rej_B2_CLdtTRdt	0.045	0.039	0.038
rej_B2_CLdtTRst	0.056	0.048	0.045
rej_B2_con1	0.052	0.053	0.059
rej_B2_con2	0.061	0.054	0.069
Simulations	1000	1000	1000

Notes: X1, X2, X3 constructed as noise.

Figure 4: Long-run Temperatures in Districts (1981-2011)



Notes:

Table 5: Effects of Temperature on the Agricultural Economy in Districts

Panel A. ICRISAT Districts (1981-2011)				
	Log Levels			
	Price Index	Yield Index	Field Wage	Cultivated Area
	(1)	(2)	(3)	(4)
$\Delta\ T_{wmin}$	-0.289***	0.174**	0.195	0.119
	(0.072)	(0.086)	(0.143)	(0.083)
$\Delta\ T_{wmax}$	0.452***	-0.210	-0.455**	-0.706***
	(0.115)	(0.140)	(0.224)	(0.163)
Observations	1,191	1,211	1,115	1,243
Sample Mean	5.66	7.14	2.86	5.45
F-stat ($T_{wmin} = T_{wmax}$)	16.93	3.10	3.33	12.22
Panel B. Census Districts (1981-2011)				
	Share of Workers			
	Cultivation	Ag. Labor	Seasonal	Non-Agriculture
	(1)	(2)	(3)	(4)
$\Delta\ T_{wmin}$	-0.010	0.082***	-0.076***	0.006
	(0.023)	(0.017)	(0.020)	(0.021)
$\Delta\ T_{wmax}$	-0.029	-0.078***	0.088***	0.016
	(0.034)	(0.025)	(0.033)	(0.035)
Observations	1,186	1,186	1,186	1,186
Sample Mean	0.32	0.18	0.17	0.32
F-stat ($T_{wmin} = T_{wmax}$)	0.12	16.15	10.27	0.03
Rainfall Control	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y

Notes: Panel A measures the price yields index for key cereals in India, weighted by their baseline (1966-1971) share of land use. Cereals: (rice, wheat, barley, maize, millets, sorghum). Panel B measures the sectoral share of workers out of total district workers (Census). Temperatures constructed as an average of the past decade's growing seasons (June-Feb). All specifications control for total precipitation in meters over the same period. The data is a decadal panel (1981-2011), using 2011 Census-consistent district boundaries. Additional controls include district and decade two-way fixed effects, fitting a state × decade linear trend. Standard errors clustered at the district level.

Figure 5: 10-Year Effects of Temperature on Urban Employment Shares

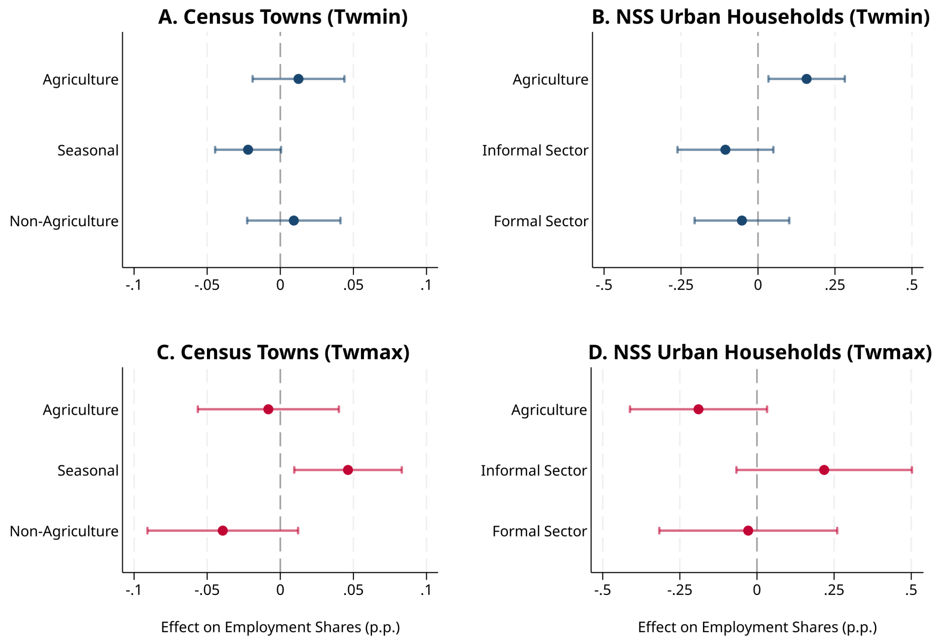
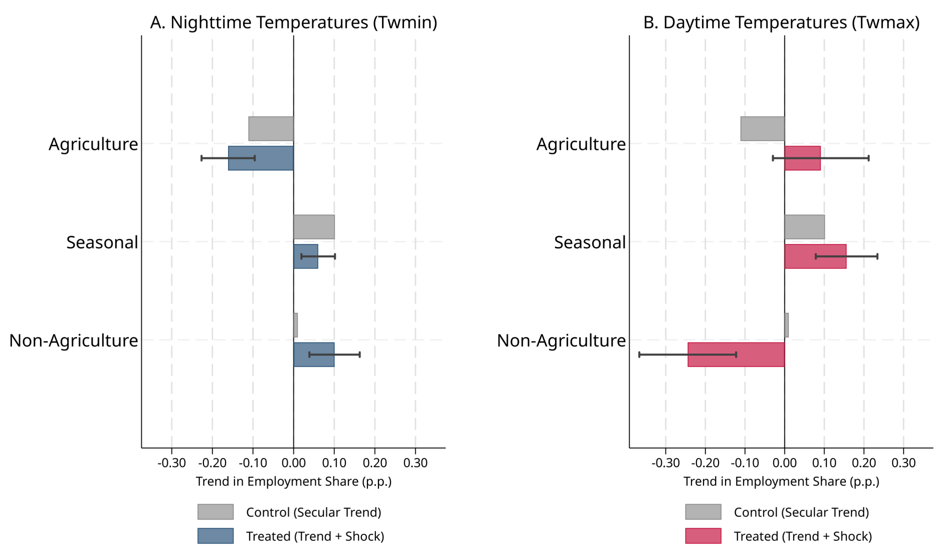


Figure 6: 30-Year Effects of Temperature on Structural Transformation in Census Towns



Notes: Figures 1 and 2 plot coefficients with 95% C.I. of rising 10-year T_{wmin} vs T_{wmax} on the share of employment per sector. Figure 1 data is a panel (Census 1981-2011; NSS 1987-2009). Figure 2 data is long-differenced Census towns. Temperatures constructed as a moving average during growing seasons (June-Feb) of the decade preceding observation. All specifications control for total precipitation in meters over the same period. Additional controls include district and decade two-way fixed effects, fitting a state $\times$ decade linear trend (state FE for the long-differenced specification). Standard errors clustered at the district level.